

Yucheng (Cecilia) Zhang

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EDUCATION

Northwestern University

M.S. in Computer Science

Aug. 2026 – May 2028

Evanston, IL

The University of Texas at Austin

B.S. in Mathematics, Minor in Computer Science | GPA: 3.74/4.00

Aug. 2023 – May 2026

Austin, TX

Relevant Coursework: Stochastic Process, Probability and Statistics, Real Analysis, Data Analytics, Database Systems, Artificial Intelligence

TECHNICAL SKILLS

Analytics/Modeling: Credit risk, risk modeling, predictive modeling, statistical analysis, feature engineering, backtesting, ROC/AUC, data validation

Languages/Tools: Python, SQL, R, MATLAB, SAS, Power BI, Git, advanced Excel reconciliation

Data Platforms: SQL Server, Oracle, PostgreSQL, Azure Synapse, Databricks, Delta Lake, AWS, Docker

Finance Domains: Credit risk, fraud detection, AML compliance, financial report analysis, SEC filings, customer risk scoring

QUANTITATIVE ANALYTICS AND FINANCIAL DATA EXPERIENCE

Enterprise Risk Analyst Intern

Toyota Industries Commercial Finance | Python, SQL Server, Azure, Power BI

May 2025 – Dec. 2025

Dallas, TX

- Built an end-to-end credit risk modeling pipeline from raw SQL Server data, including ETL, feature engineering, Logistic Regression, and Random Forest training.
- Engineered credit risk features including PayNet inquiries, loan-to-value ratios, account tenure, asset segments, and loan servicing attributes for trucks, trailers, and used-vehicle portfolios.
- Validated model performance with ROC/AUC, confusion matrices, and feature importance, achieving AUC > 0.82 and improving risk-flagging accuracy by 3–4%.
- Backtested predicted risk buckets against observed outcomes, identifying up to 3.7× higher default rates in high-risk groups and informing credit scorecard segmentation refinements.
- Automated monthly model refreshes with parameterized SQL and Python workflows connected to Power BI dashboards, reducing manual reporting effort by 10% and enabling real-time risk tracking.

Financial Data Development Engineer – Fraud Detection

Caitong Securities Ltd | SQL, Oracle, Stored Procedures

Jul. 2024 – Aug. 2024

Hangzhou, China

- Designed SQL ETL pipelines processing 10M+ daily transactions with window functions, subqueries, and stored procedures, improving extraction efficiency by 5%.
- Implemented automated fraud-detection logic with parameterized thresholds and modular stored procedures, reducing false positives by 2% and improving reliability.
- Developed a customer risk scoring engine by integrating transactional and behavioral data streams, enhancing fraud detection precision and supporting AML compliance.

FinChat – AI-Powered Financial Analysis System

Southern Methodist University | Python, FinGPT, LangChain, CrewAI

Jan. 2023 – May 2024

Remote

- Built ETL pipelines to extract and process 100+ SEC filings from EDGAR, structuring unstructured financial data into a scalable data lake for metric extraction and trend detection.
- Implemented LangChain and CrewAI workflows for contextual financial interpretation, improving data processing accuracy by 3% and reducing manual analysis time.
- Collaborated with a cross-university team to support LLM-based financial report analysis for scalable analysis of company filings.

Software Engineering Intern – Customer Analytics

Joblogic-X Corporation | Python, APIs, Git

Nov. 2023 – Jun. 2024

Dallas, TX

- Integrated social-media and transactional order data into a scalable customer analytics pipeline for data-driven marketing and engagement strategy.
- Implemented Python ranking and clustering algorithms, producing prioritized customer lists and reproducible analytics outputs for downstream chatbot workflows.

ADDITIONAL QUANTITATIVE RESEARCH

Researcher & TA – Biological Data Analytics

University of Texas at Austin | R, Linux, AWS, Docker

Jan. 2024 – Dec. 2024

Austin, TX

- Processed 250GB of RNA-seq data, performed Voom/Limma differential-expression analysis, identified 48 key genes, and automated PCA/volcano-plot workflows in R.
- Applied reproducible data pipelines and statistical analysis across large datasets, improving processing speed by 6% and interpretation efficiency by 15%.

SELECTED PUBLICATIONS

- Coauthored 5 publications across machine learning, neuroengineering, computer vision, and social media analysis, including *Expert Systems with Applications*, *Frontiers in Neurology*, *Frontiers in Neuroscience*, and *IEEE ICIPCA*.